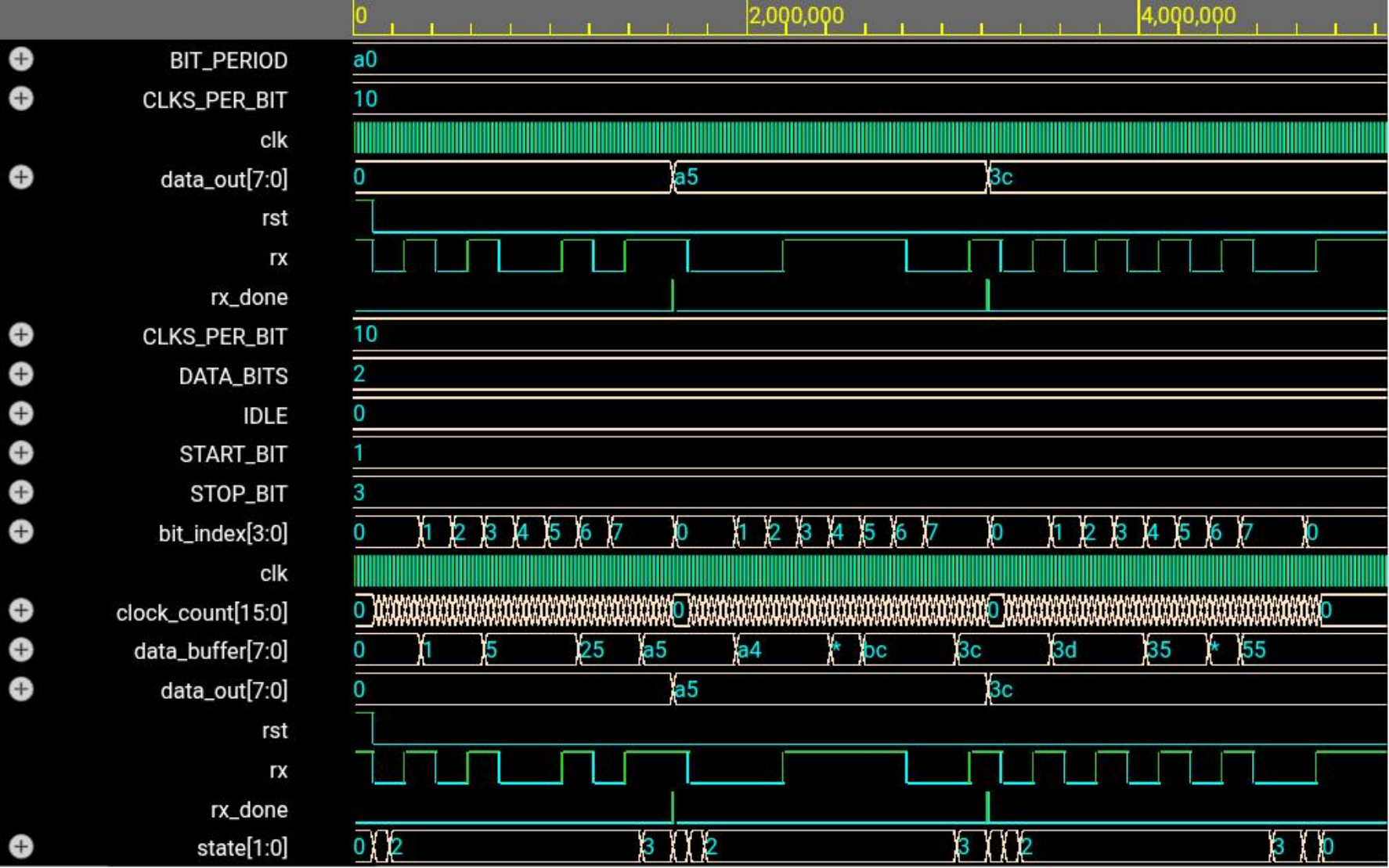
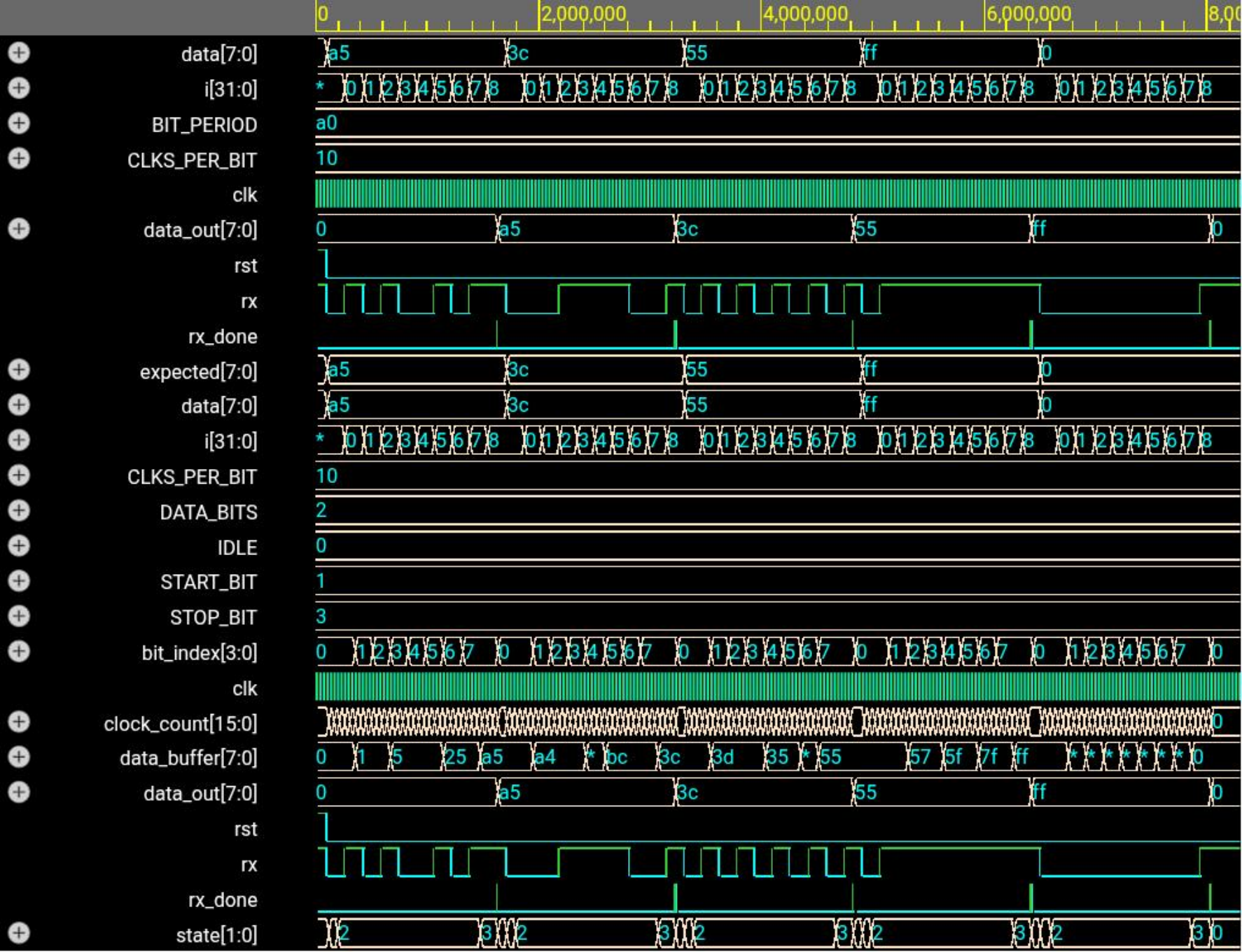

0
2,000,000
4,000,000
BIT_PERIOD
a0
CLKS_PER_BIT
10
clk
data_out[7:0]
0
a5
3c
rst
rx
rx_done
CLKS_PER_BIT
10
DATA_BITS
2
IDLE
0
START_BIT
1
STOP_BIT
3
bit_index[3:0]
0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 0
clk
clock_count[15:0]
0
0
0
0
data_buffer[7:0]
0 1 5 25 a5 a4 * bc 3c 3d 35 * 55
data_out[7:0]
0
a5
3c
rst
rx
rx_done
state[1:0]
0 2 3 2 3 2 3 0


0
2,000,000
4,000,000
6,000,000
data[7:0]
i[31:0]
BIT_PERIOD
CLKS_PER_BIT
clk
data_out[7:0]
rst
rx
rx_done
expected[7:0]
data[7:0]
i[31:0]
CLKS_PER_BIT
DATA_BITS
IDLE
START_BIT
STOP_BIT
bit_index[3:0]
clk
clock_count[15:0]
data_buffer[7:0]
data_out[7:0]
rst
rx
rx_done
state[1:0]